

Computer Assisted Proofs

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Introduction

A big spectrum

pencil and paper proofs

⋮

formally verified proofs

own computer programs

⋮

automatic theorems provers

⋮

interactive theorems provers

Some families in the spectrum

- Automatic theorems provers for first-order logic
- Satisfiability modulo theories solvers
- Inductive theorem provers
- Interactive proof assistants
- Tools for the verification of programs

Automatic Theorems Provers for First-Order Logic

Some features

- The TPTP World

“A well know and established infrastructure that supports research, development, and deployment of automated theorem proving systems for classical logics.” (Sutcliffe 2010, p. 1.)

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The World Championship for Automated Theorem Proving!

Automatic Theorems Provers for First-Order Logic (ATPs)

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- Some ATPs for (classic) first-order logic: E, Equinox, Metis, SPASS and Vampire.

History

- **FOF** (Full First-Order Form): Full first-order logic (Sutcliffe 2009).
- **CNF** (Clause Normal Form): First-order logic in conjunctive clausal form (Sutcliffe 2009).
- **TFF** (Typed First-Order Form): **TF0** (monomorphic) (Sutcliffe, Schulz, Claessen and Baumgartner 2012) and **TF1** (rank-1 polymorphism) (Blanchette and Paskevich 2013).
- **THF** (Typed Higher-Order Form): **TH0** (monomorphic) (Sutcliffe and Benz Müller 2010) and **TH1** (rank-1 polymorphism) (Kaliszyk, Sutcliffe, and Rabe 2016).
- **TXF** (Typed Extended First-Order Form): **TX0** (monomorphic) and **TX1** (rank-1 polymorphism) (Sutcliffe and Kotelnikov 2018).

Description

`fof(name, role, formula)` where:

`name` identifies the formula,

`formula` is a FOL-formula and

`role` $\in \{ \text{axiom, definition, hypothesis, conjecture} \}$.

Each TPTP problem contains one or more annotated formulae.

Logical constant	Symbol
True	\$true
False	\$false
Conjunction	&
Disjunction	
Negation	~
Conditional	=>
Biconditional	<=>
Universal quantifier	! [X] : p(X)
Existential quantifier	? [X] : p(X)

Satisfiability Modulo Theories Solvers

Satisfiability Modulo Theories

- The study of automatic methods for checking the satisfiability of first-order formulae with respect to some **background** theory is called Satisfiability Modulo Theories (SMT) (Barret, Sebastiani, Seshia and Tinelli 2009).

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Theories of real numbers, integers, arrays, lists, records, bit-vectors,...

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SMT-LIB

The SMT library and standard.

Some SMT solvers

- Z3 (Microsoft), CVC4 and veriT.

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Z3 demo

See **Z3** home page.

Inductive Theorem Provers

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- ITPs (Google for List of All Inductive Theorem Provers in The World).

Interactive Proof Assistants

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*“Proof assistants are computer systems that allow a user to do mathematics on a computer, but not so much the computing (numerical or symbolical) aspect of mathematics but the aspects of **proving** and **defining**. So a user can **set up** a mathematical theory, define properties and do logical reasoning with them.” (Geuvers 2009, p. 3.)*

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Example

- Based on set theory: **Isabelle/ZFC**, **Metamath** and **Mizar**
- Based on higher-order logic: **HOL4**, **HOL Light** and **Isabelle/HOL**
- Bases on type theories: **Agda**, **Rcoq** and **Lean**.

Some Proof Assistants

- Agda
- Isabelle
- Lean
- Rcoq

- Chalmers University of Technology and University of Gothenburg (Sweden)
- Based on Martin-Löf Type Theory
- Direct manipulation of proofs-objects
- Back-ends to **Haskell** (**GHC**) and **JavaScript**
- Written in **Haskell**

- University of Cambridge (England) and Technical University of Munich (German)
- Based on HOL (Higher Order Logic)
- Tactic-based
- Extraction of programs to **Haskell**, **OCaml**, **Scala** and **SML**
- Written in **SML**
- Integration with ATPs and SMT solvers

- Microsoft Research and Carnegie Mellon University (USA)
- Based on dependent type theory
- Declarative style and tactic-based
- Written in C++
- Integration with ATPs and SMT solvers

- INRIA (France)
- Based on the Calculus of Inductive Constructions
- Tactic-based
- Extraction of programs to Haskell, OCaml and Scheme
- Written in OCaml (with a bit of C)

Real Applications

1977 PhD thesis (van Benthem Jutting 1977) ([Automath](#)).

One of the largest repository of formalised mathematics (1990–current).

MML includes (Bancerek, Byliński, Grabowski, Korniłowicz, Matuszewski, Naumowicz and Pak 2018):

- over 12000 definitions
- over 59000 theorems
- over 1290 articles
- over 253 authors

Proof of the Four Colour Theorem

- 1976** Every planar graph is four colorable. Announce in the Bulletin of AMS (K. Appel and Haken 1976).
- 1977** Description of the proof in Scientific American (K. Appel and Haken 1977).
- 1977** Simpler proof (still using a computer) (Robertson, Sanders, Seymour and Thomas 1997).
- 2005** Formalised proof. Microsoft report (Gonthier 2005) (Rcoq).
- 2008** Description of the formalised proof in Notices of AMS (Gonthier 2008).

2006 Journal (and proceedings) publication: POPL 2006 ([Rcoq](#)).

2009 Description in CACM (Leroy [2009](#)).

2009 Proceedings of SOSP 2009 ([Isabelle/HOL](#)).

2010 Description in CACM (Klein, Andronick, Elphinstone, Heiser, Cock, Derrin, Elkaduwe, Engelhardt, Kolanski, Norrish, T. Sewell, Tuch and Winwood [2010](#)).

Proof of the Odd-Order Theorem

- 1963** Solvability of groups of odd-order (Feit and Thompson 1963).
- 2012** Announce of the formalised proof (Gonthier and Théry 2012) (Rcoq).
- 2013** Formalised proof. Journal publication (Gonthier, Asperti, Avigad, Bertot, Cohen, Garillot, Le Roux, Mahboubi, O'Connor, Biha, Pasca, Rideau, Solovyev, Tassi and Théry 2013).

Proof of Kepler Conjecture

1998 Hales announced proof of Kepler Conjecture.

2005 Journal publication “uncertified” (Hales 2005).

2014 Announce of the formalised proof (Hales 2014) (*Isabelle* and *Isabelle/HOL*, *HOL Light*, *OCaml*, *CamLP5* and *GLPK*).

2017 Formalised proof. Journal publication (Hales, Adams, Bauer, Dang, Harrison, Hoang, Kaliszyk, Magron, McLaughlin, T. T. Nguyen, Q. T. Nguyen, Nipkow, Steven, Pleso, Rute, Solovyev, Ta, Tran, Trieu, Urban, Vu and Zumkeller 2017).

2014 Journal publication (Marché 2014) (Frama-C).

2014 Journal publication (Kumar, Myreen, Norrish and Owens 2014) (HOL4).

2015 Journal publication (A. W. Appel 2015) (Rcoq).

2015 Proceedings of SOSP 2015 ([Rcoq](#)).

2017 Description in CACM (Chajed, Chen, Chlipala, Kaashoek, Zeldovich and Ziegler [2017](#)).

Rigorous Test-Oracle Specification and Validation for TCP/IP and the Sockets API

2001 Technical report ([HOL4](#)).

2015 Proceedings publication (2008, 2006, 2005, 2002, 2001).

2017 Journal publication (Bishop, Fairbairn, Mehnert, Norrish, Ridge, P. Sewell, Smith and Wansbrough [2018](#)).

2019 Proceedings of PLDI 2019 (Löow, Kumar, Tan, Myreen, Norrish, Abrahamsson and Fox [2019](#)) ([HOL4](#)).

2019 Journal publication (Beeson, Narboux and Wiedijk 2019) (HOL Light and Rcoq).

2022 Description in The American Mathematical Monthly (Beeson 2022).

Resolution of Keller's Conjecture

2020 Proceedings of IJCAR 2020 (SMT solver).

2021 Description in CACM (Monroe [2021](#)).

2020 Proceedings of IJCAR 2020 (*Isabelle/HOL*).

2022 Journal publication (Eßmann, Nipkow, Robillard and Sulejmani *2022*).

Tools for the Verification of Programs

Dedicate Tools and Systems for the Verification of Functional Programs

- Hip: The Haskell Inductive Prover

Dedicate Tools and Systems for the Verification of Functional Programs

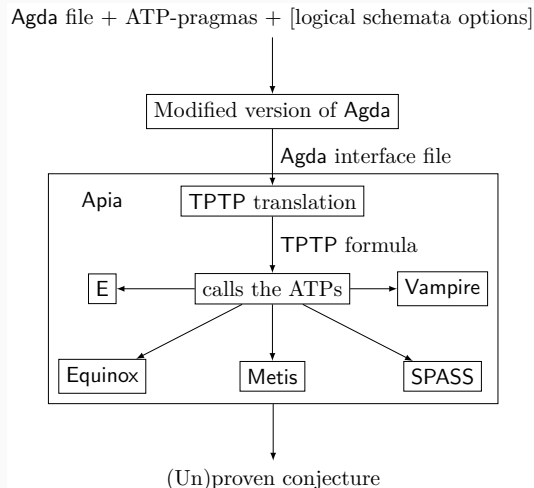
- **Hip**: The Haskell Inductive Prover
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Dedicate Tools and Systems for the Verification of Functional Programs

- **Hip**: The Haskell Inductive Prover
- **Sparkle**: An interactive proof assistant for Clean
- **Zeno**: An automated proof system for Haskell program properties

An Application: Reasoning about Functional Programs

Combining Agda with ATPs*



*Figure source: Sicard-Ramírez (2015, Fig. 6.1).

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